

AI-Driven Breed Classification and Smart Trading Platform for Cattle and Buffaloes

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Abstract - LivestockIQ is an AI-powered web-based system used to automate livestock classification, disease prediction, and a trading platform using image processing and ML techniques. Traditional methods depend on manual processes, which consume more time and lead to errors by humans. This system solves these challenges by providing an AI-based solution that provides accurate classification and health checks through image analysis. This system provides user authentication and role-based access for buyers and sellers. Sellers upload the images of cattle or buffaloes by providing the details like price, location, etc. The uploaded images are preprocessed using computer vision to obtain visual features and analyzed using ML models. The system also predicts diseases and provides basic treatments and recommendations. Buyers can visit the website and search for cattle or buffaloes based on breed and seller verifications. This project provides a digital marketplace, that ensures fair trading and communication between buyers and sellers. LivestockIQ reduces manual work and promoting a smart digital trading platform and used in rural areas and other initiatives taken by the government or government aided organizations.

Keywords – Image classification, Disease prediction, Machine Learning, Smart trading platform

I. INTRODUCTION

LivestockIQ plays an important role in the agricultural sector by supporting dairy products like milk, butter,

etc. In countries like India, cattle and buffaloes are important in the dairy industry, so their breed quality, structure, health are important factors. Traditional methods rely on manual inspection done by trainers. These methods are time consuming and sometimes leads to human error. Livestock trading is mostly done through unorganized channels or medium which leads to risk of fraud and unhealthy animal sales. No existing system combines breed classification, disease prediction and smart trading platform in a single platform or product – this is where LivestockIQ solves the problem. In recent days, AI, ML, and image processing solve these challenges. It improves accuracy and consistency by making an objective evaluation. It promotes livestock trading a better smart digital marketplace combining all these functionalities. This paper presents LivestockIQ that combines image based classification, disease prediction and buyer-seller trading platform. This system allow sellers to upload their animal images to the system for preprocessing, visualizing, classifying and prediction. The outcomes are kept in the record with the additional details about the seller. Buyers look into the system and the stored results are shown in the user interface. They can contact with the seller and make a fair trade. LivestockIQ's main aim is to improve livestock evaluation, reduce manual workload, and making a smart digital platform.

II. LITERATURE SURVEY

Several studies provided the routine of Artificial Intelligence, Machine Learning and Image Processing. These studies focus on breed classification, disease prediction and agriculture systems. Early studies rely on manual measuring tools and scoring systems. These approaches helps in domain knowledge.

Using artificial intelligence (AI), specifically the cutting-edge InceptionResNetV2 architecture inside a CNN algorithm, this system can distinguish between buffalo and cows with remarkable accuracy. Its current use case is identifying cattle, but it may easily be extended to encompass insurance claim evaluation and cow healthcare. To train and validate, the system uses a dataset that contains high-resolution photographs of cattle from various geographies and habitats [1].

This study compares the outcomes of using the Hybrid Real Coded Genetic Algorithm (RGA) to those of using the RGA with various versions, including RGA without mutation, RGA without crossover, and RGA with mutation and crossover, to solve a direct and penalty area programming model for pre-natal dairy buffalo. Other nutritional models, such as CNCPS and INRA, can also be used with these models. This book is a great first step in the right direction because it gives a strong and exact standard for other researchers in the field of animal food and gives the groundwork for building an Android app and software to frame the least-cost portion based on the nutrient supplies of cattle and buffalo, which have been determined according to the Indian Council of Agricultural Research (ICAR, 2013) and the National Research Council (NRC, 2001) on a dry matter basis. Using the information and algorithms presented in this book, dairy farmers are able to provide their animals with a healthy, reasonably priced food [2].

The CNN algorithm, built using the InceptionResNetV2 architecture, sorts cattle according to characteristics including horn form, body size, coat color, and face features. Additionally, it uses the YOLOv8 muzzle point recognition method to uniquely identify each cow. This enables the exact monitoring and control of specific livestock. Users can also make use of the system's similarity comparisons to see how their input cattle photos stack up against the database of recognized cattle. To determine the degree of resemblance and to identify individual animals, it employs state-of-the-art image recognition algorithms [3].

Based on their experimental examination, this article also examines the methods and instruments. In this article, we summarize the system's routine in terms of

recognition accuracy over the many procedures after testing them on various datasets to determine how well they identified the cattle. In this post, we have attempted to cover every possible benefit of cattle identification systems. The authors also noted the importance of creating a reliable method for livestock identification. Additionally, this page provides guidance for further research in this specific area. Particularly concerning is the lack of research on the use of muzzle points for livestock identification; further investigation into this area is necessary [4].

Connected to the Internet of Things (IoT), it utilizes deep learning algorithms for early disease detection, behavior analysis, and feeding system optimization; furthermore, it enables low-latency decision-making at the farm level. In keeping with the One Health principle, the framework takes into account the interconnectedness of animal welfare, ecological sustainability, and food security. To evaluate the model's functionality in a practical setting, a dairy and poultry farm were set up on the field for experiments. The proposed system could detect early signs of common livestock diseases like mastitis and lameness with a 94% accuracy rate, improve feed efficiency by 15%, and reduce methane emissions by 10% through intelligent ventilation and waste control [5].

This work introduces the first robotic milking system that combines artificial intelligence-driven thermal imaging with actual, non-aggressive nursing of udder fitness in Italian Mediterranean buffalo. The method uses a calibrated weather station to account for climatic variables and takes advantage of the matched attainment of thermal pictures in milking, which is different from past systems [6].

The livestock sector could use a simpler way to acquire and sell animals like goats, chickens, and calves; therefore, the idea behind Enhancing Livestock Management through Innovative Solutions for Effective Trading Practices is to provide a user-friendly online platform. The platform facilitates transactions between buyers and sellers with an emphasis on security, transparency, and efficiency. Important functions include keeping tabs on medical history, notifying users of immunization or breeding dates, and displaying the current availability of animals on the platform [7].

The cattle sector can significantly contribute to solving these problems. Sustainable production has long been the center of attention, but the United Nations' 2030 Agenda has shifted the focus to the sector's role in attaining the SDGs. Insightful and succinct, this essay summarizes the successes and failures of the Indian

livestock sector's transition to sustainable growth. To guarantee a prosperous cattle sector in India, policymakers, academics, and stakeholders can use these insights to push for the adoption of sustainable methods [8].

However, they suffered inconsistency and highly dependent on manual workload. To overcome these limitations, researchers started applying computer vision techniques to analyse the animal body structure and diseases[9]. By making implementation of user logins, our application provides real world smart trading platforms. Recent studies have used CNN [10] to identify coat color, body shape, horn structure and other details. These models have improved their accuracy and solved the manual workload. Disease prediction also gained performance attention by identifying skin lesions, lameness and external infections using deep learning models. Some research says that smart agriculture platforms using IoT sensors, health monitoring devices, These systems are costly and unsuitable for small scale farmers. Unlike existing systems, LivestockIQ combines image-based breed classification, disease prediction and trading marketplace [11]. It requires minimal hardware requirements and have a practical deployment. Sellers upload the images and buyers access the verified animal information. This system joins the gap between analysis and transparent marketplace [12].

III. PROPOSED METHOD

LivestockIQ is a modular, AI-based web application that combines image based classification, disease prediction and a smart digital marketplace for buyers and sellers. The system's aim is to automate evaluation to promote transparency and ease of use. Fig. 1.0 illustrates the overall system architecture follows a role-based working where users interact as sellers or buyers. The images are analyzed by the techniques of ML and extract the features then stored in the database.

The planned method is partitioned into the next modules:

1. User authentication module:

This module handles user registration and login using security functions. The authorized users access to the system and view the dashboard based on seller or buyer.

2. Seller module:

This module allows owners or farmers to upload the image of their own cattle or buffaloes with additional details. The processed image and details are stored in the database for buyer access.

3. Image processing and feature extraction module:

In this module, the uploaded image goes on preprocessing steps. Here the removal of noise, visual quality, and features are extracted. The features are the input for image classification and disease prediction.

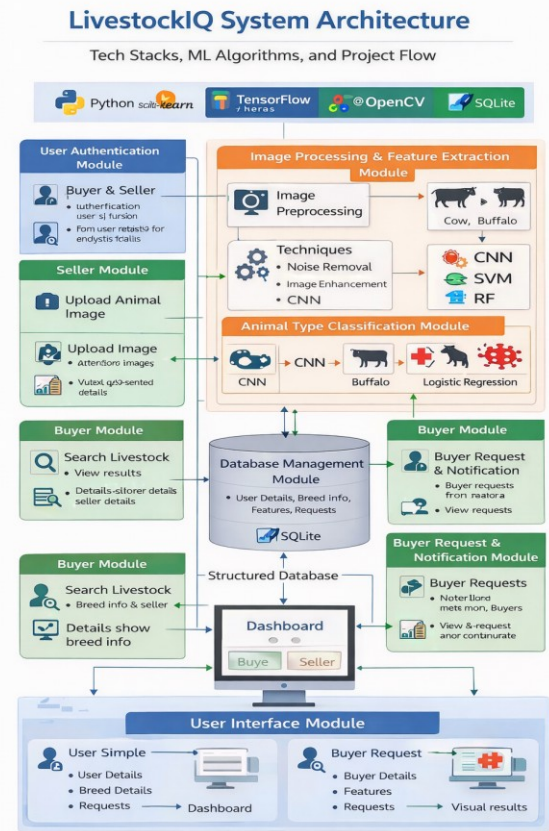


Fig. 1.0 System Architecture

4. Animal type classification module:

This module identifies the type and breed of the cattle or buffalo by using the features extracted. ML methods used to classify the images and it is displayed to the seller and thus stored in the database [13].

5. Disease prediction module:

This module analyzes the visual and external symptoms present in the image. This helps to detect health issues. If a disease is found, the system provides some remedies and suggestions for prevention. This also recorded for buyer reference.

6. Buyer module:

In this model, buyer can search for cattle or buffalo by selecting the breed. The system shows the results from the database. The breed details with seller details also shown in the interface. Then buyer can select the breed to buy and contact the seller.

7. Buyer request and notification module:

In this module, buyer shows interest in buying the animal. The request is stored in the database and the seller can view the request and contact the buyer for further communication.

8. Database management module:

The module that collects all the data such as user details, breed details, features, requests, etc. and stored in the database. Structured database is used here to ensure consistency and efficiency [15].

9. User interface module:

The user interface module delivers a basic user interface for users (buyers and sellers). The design is easy for navigation, usage and clear display of results.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

The LivestockIQ system was assessed its performance in animal type differentiation, disease prediction, and overall system performance. Fig. 1.1 describes the performance of the various ML models. The accuracy of the datasets are measured and visualized using confusion matrix metrics. The experiments were done using a dataset collected from publicly available sources and locally captured images. The dataset included multiple breed data that took in multiple light conditions.

1. Experimental setup

The system was implemented using Python for image preprocessing and machine learning integration. The dataset contained about 1200 cattle and buffalo images from public datasets and local farms, divided into training (80%) and testing (20%) sets. Images were resized to 224×224 pixels and preprocessed before

classification. A CNN model was trained using TensorFlow for 25 epochs and integrated into the web application. Fig 1.2 shows the training curves representing model performance.

2. Animal type classification results:

The animal type classification module provided reliable performance in identifying cattle and buffalo breeds from the dataset. The classification model achieved an overall accuracy of 92.4% on the testing dataset, with precision and recall values of 91.8% and 92.1% respectively. The confusion matrix in Figure 1.1 illustrates the classification performance across different cattle breeds. The Figure represents the breed class distribution and differentiation and visualized using charts and graphs.

3. Disease prediction results:

The disease prediction module identifies the visible, external diseases such as skin infections, lesions, etc. The system generates alerts, recommendations to the users for preventive care [14]. The Fig. 1.4 shows the feature filtered from various CNN layers of the class Ayrshire cattle. Fig. 1.5 delivers the different patterns of the edges and shapes of the Brown Swiss cattle. Fig. 1.6 and Fig. 1.7 shows the Holstein Friesian cattle and Jersey cattle respectively which shows the various breeds classification images and its edges. So each breed varies in its structure, size, shape and performance is based on each breed type and analyzed based on that breed type. The module performed well for visually detectable diseases and limited to constraints like internal or non-internal conditions.

4. Buyer-seller interaction performance:

Fig. 1.8 show the final result of the experiments. This Figure shows how the breed is preprocessed, trained and tested, and based on that, the results will be generated. The buyer seller workflow was tested for accuracy and response time. Buyer workflow based on breed selection that retrieved from the database. The seller adds the image that preprocesses, undergoes machine learning classification and prediction and stores in the database [16]. This project adds security and privacy concerns using authentication and authorization, so that fraudulent transactions and unauthorized communications will be prevented, thereby improving the scalability and reliability across all areas.

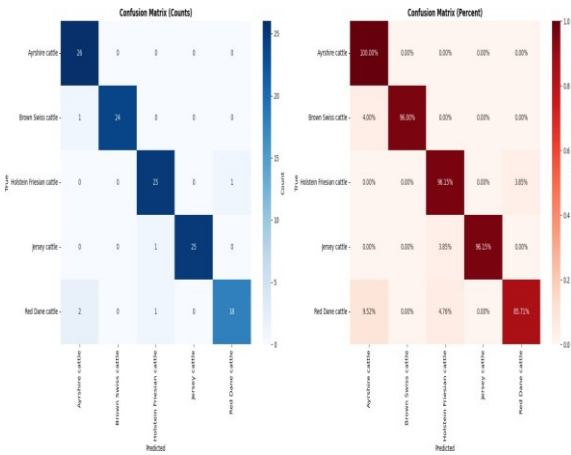


Fig. 1.1 Confusion Matrix

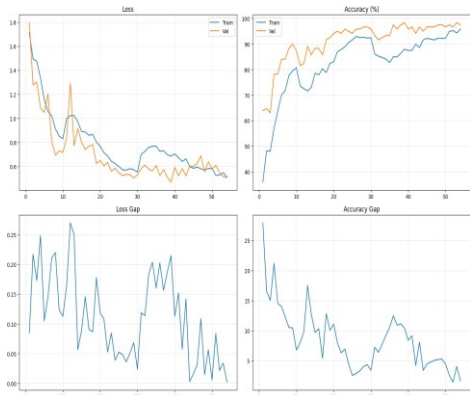


Fig. 1.2 Training Curves

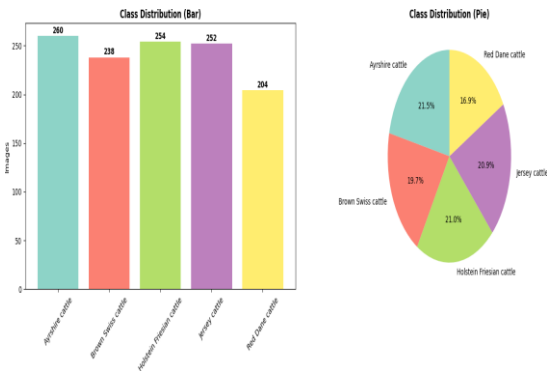


Fig. 1.3 Class Distribution

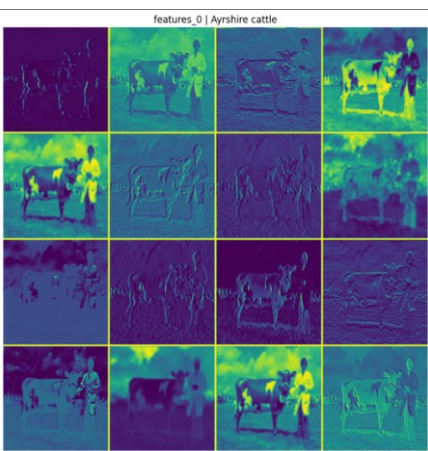


Fig. 1.4 Image Processing – I

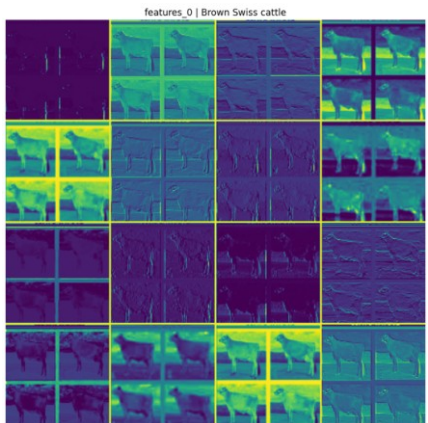


Fig. 1.5 Image Processing – II

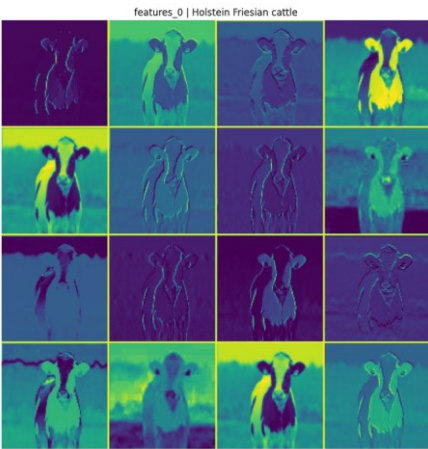


Fig. 1.6 Image Processing - III

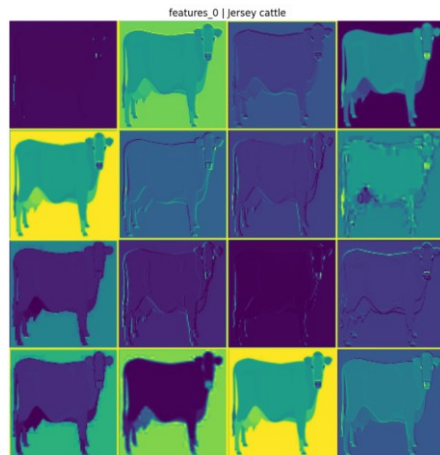


Fig. 1.7 Image Processing - IV

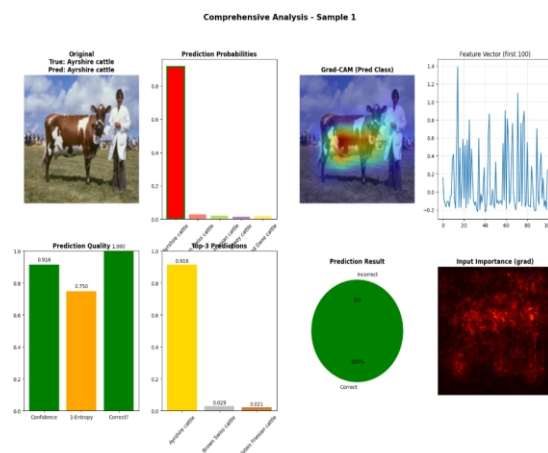


Fig. 1.8 Final Results

5. Discussion:

To evaluate the effectiveness of the proposed LivestockIQ system, its performance was compared with traditional machine learning methods such as Support Vector Machine (SVM) and Random Forest. The CNN-based model achieved higher accuracy (92.4%) compared to SVM (84%) and Random Forest (87%), demonstrating improved performance in livestock breed classification.

6. Summary of Results:

The experimental results of LivestockIQ shows that the classification and disease prediction within the trade platform. The system achieved reliable results in classification performance with the conditions of

standard good images. Buyer requests and seller data were processed so that the interaction is effective between the users. Overall, the results demonstrated that this system provides a real-world, accurate and accessible solution for livestock and digital platform.

V. CONCLUSION

This paper presents LivestockIQ, an AI based web application system that integrates breed classification, disease prediction and a smart digital trading platform for buffaloes and cattles. Traditional methods rely on manual inspection and checks and have an informal marketing platforms and channels, that lead to human errors, lacking transparency and reducing efficiencies. The proposed system solves these problems by using machine learning and image processing to automate livestock analysis and providing a reliable marketplace for buyers and sellers. The developed platform includes modules such as user authentication, image processing, feature extraction, classification and disease prediction, buyer seller interaction and a secure communication channel. Experimental evaluations and tests was conducted using dataset collected from public sources and images captured locally. The ML model performs effective classification by performing feature extraction to find inner details of the breed and classify accordingly, and in turn it predicts diseases also. By combining AI and digital marketplace, the system provides a modern way of livestock management supporting farmers and traders. Overall, the system shows a practical, modern solution for livestock management. Future enhancements may be added with the advance deep learning models, expanding datasets, integrating with IoT based monitoring systems and integration with mobile applications to improve scalability and reliability and improvement in performance.

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